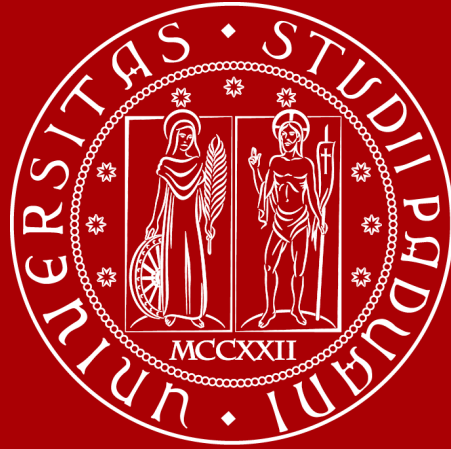




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Master thesis proposal

Presented By:

Daniele Zebebe



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COLLABORATION



Istituto Nazionale di Fisica Nucleare



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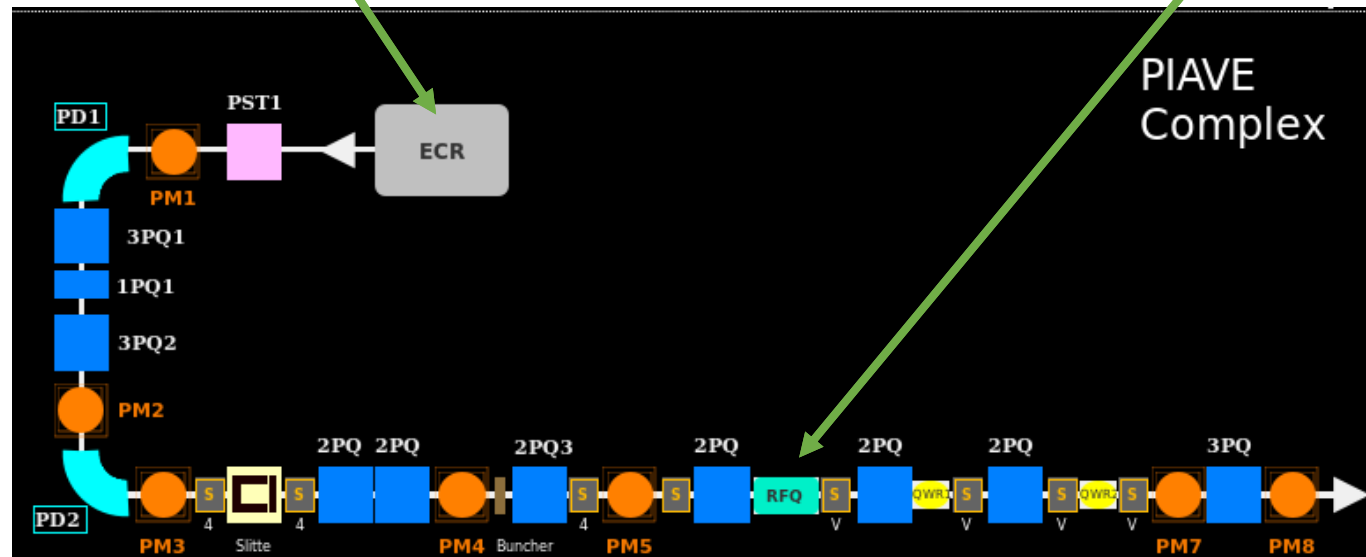
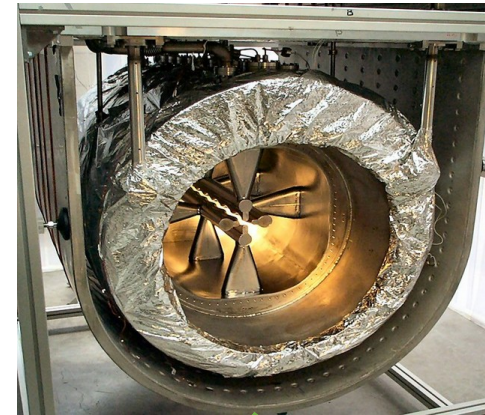
ACCELERATOR

BEAM'S ACCELERATION

ECR
(Electron
Cyclotron
Resonance)



RFQ
(Radio-
Frequency
Quadrupole)

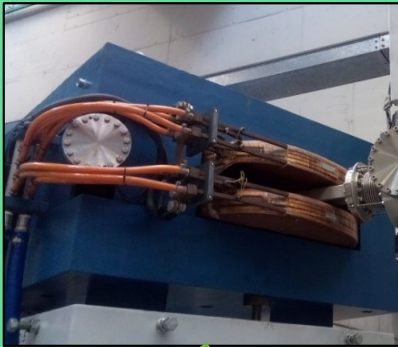




ACCELERATOR

BEAM'S CONTROL

Dipole



Quadrupole

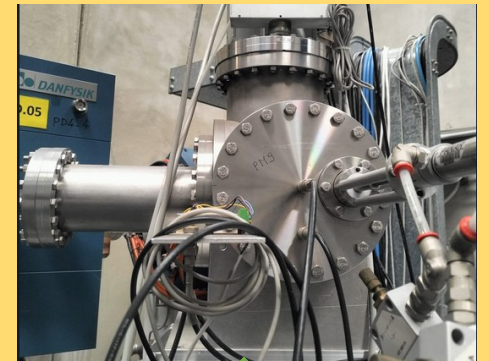


Steerer

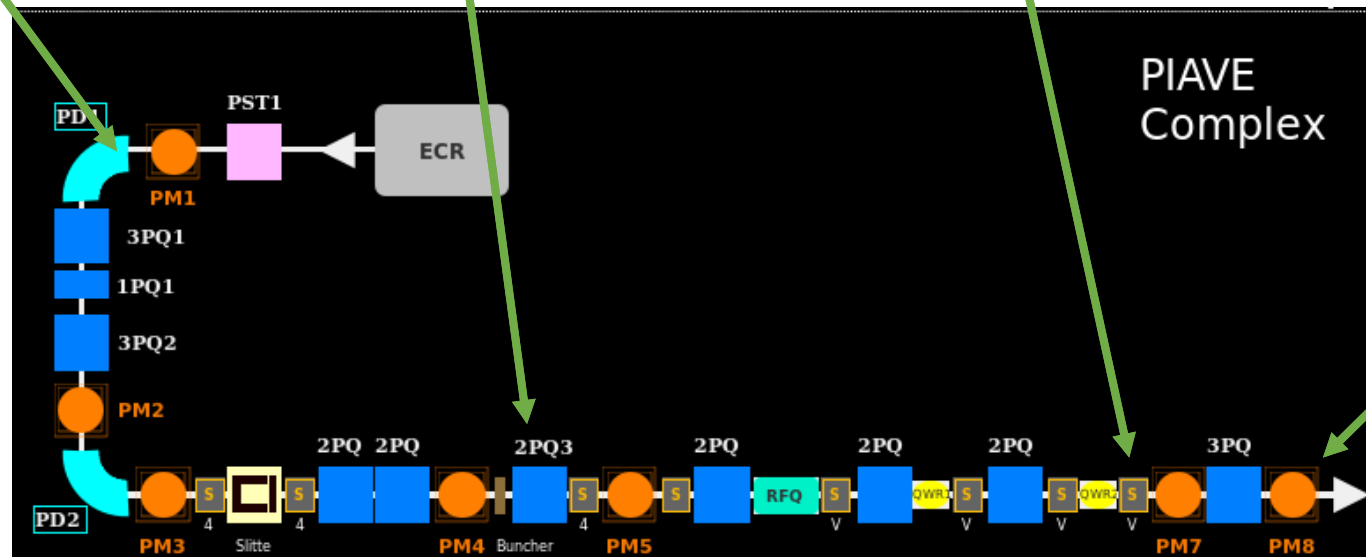


Beam
Transport
System

Diagnostic Box
(XY Grids and
Faraday Cup)



Each module
can be set and
it influences
the beam's
quality



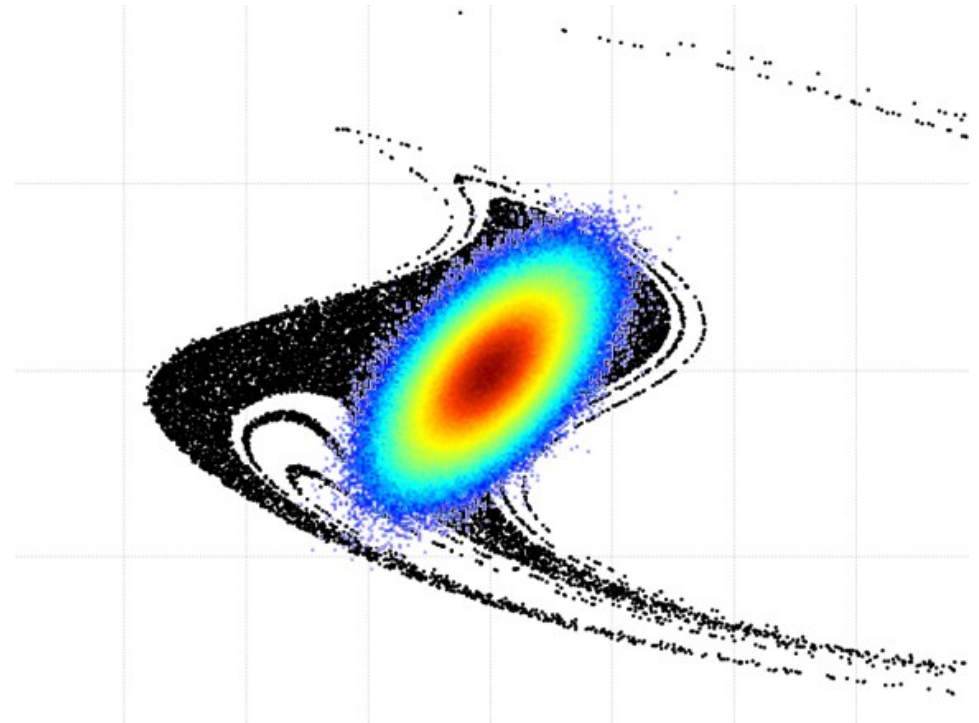


BEAM EMITTANCE

Measures the quality of the beam.

It should be:

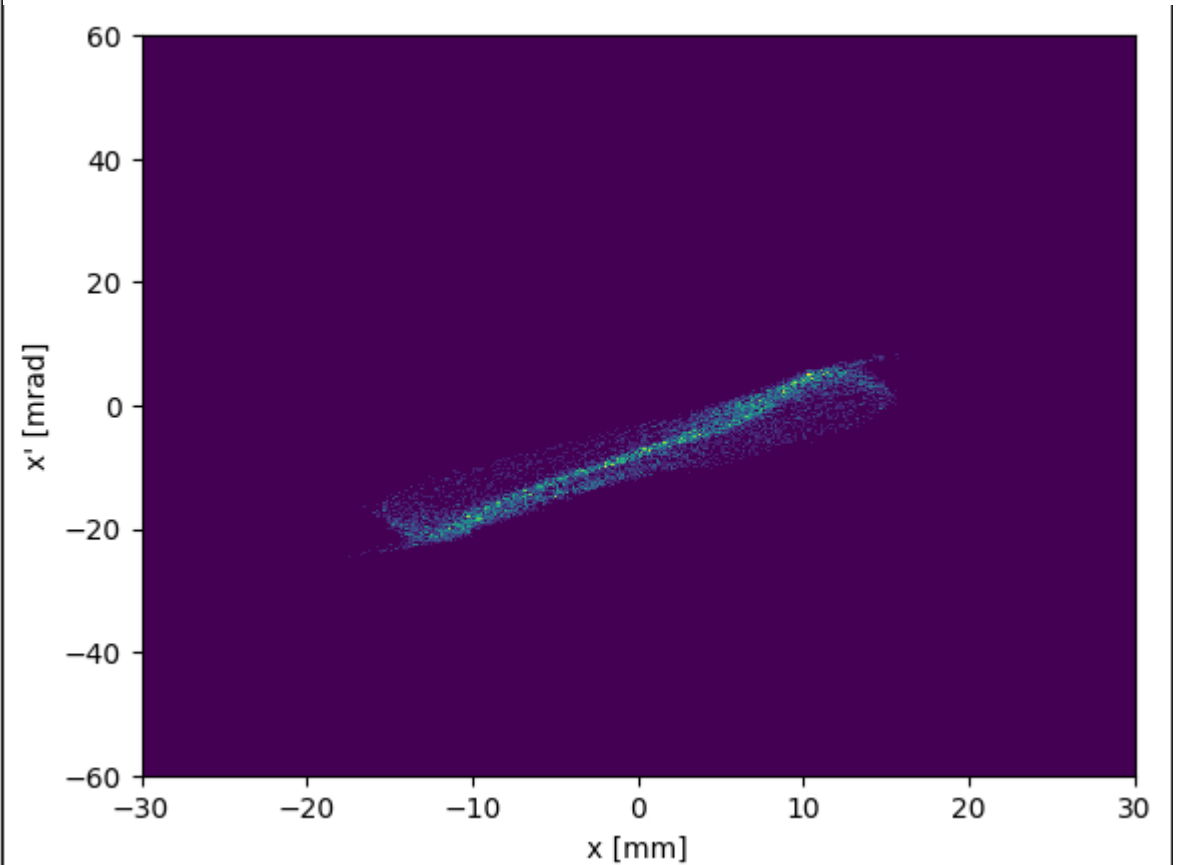
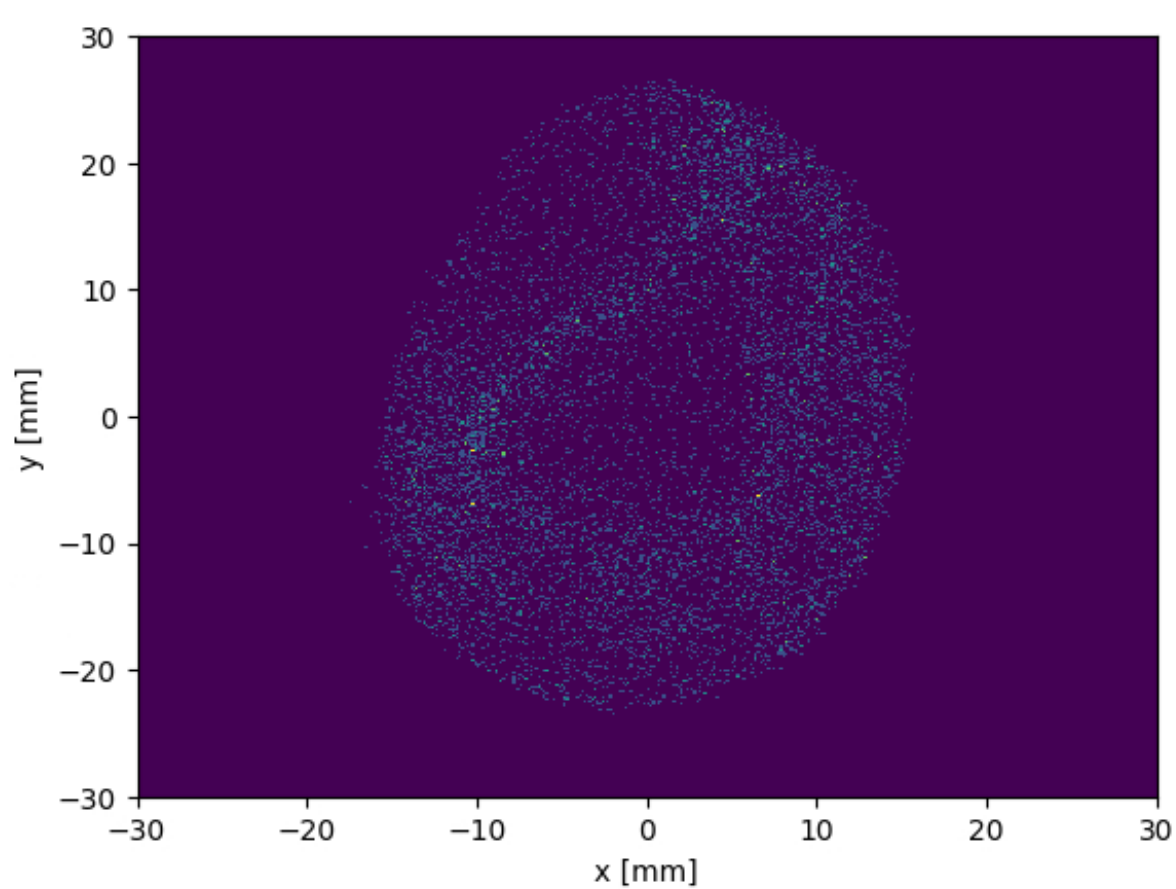
- Intense
- elliptical





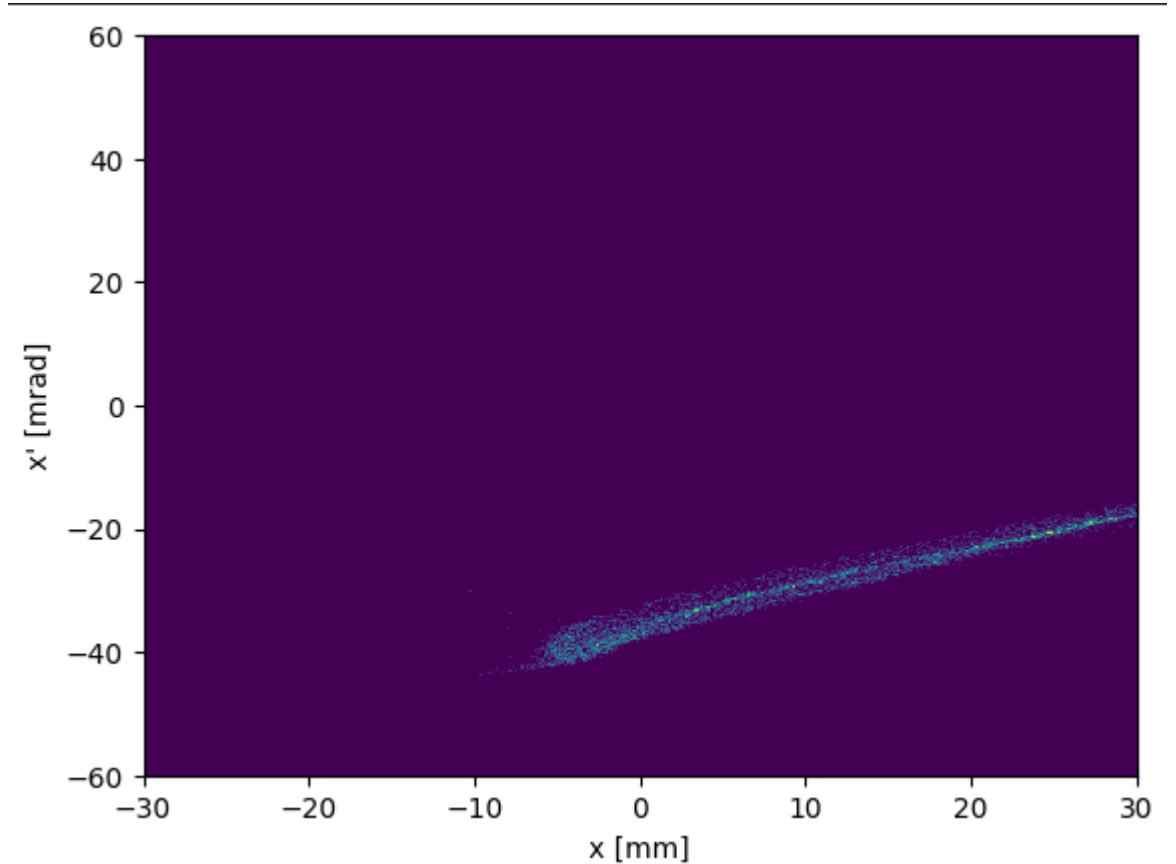
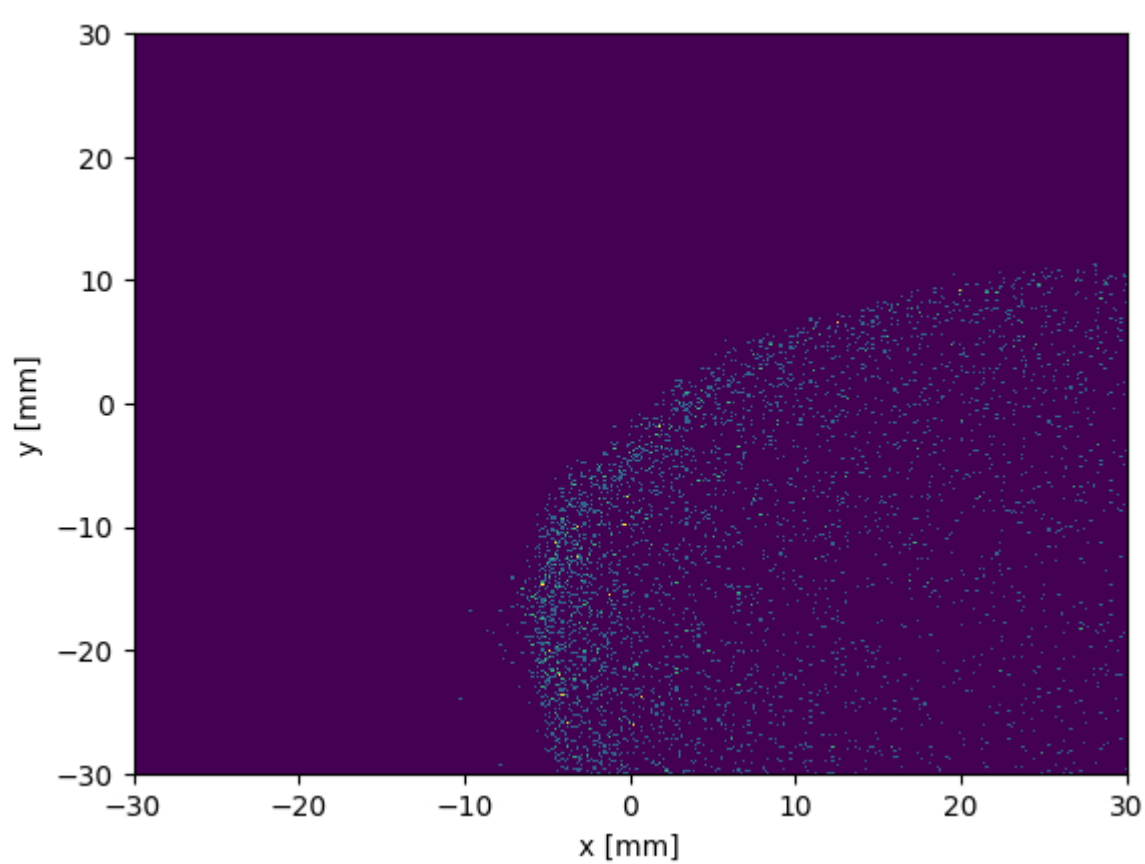
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BEAM EMITTANCE





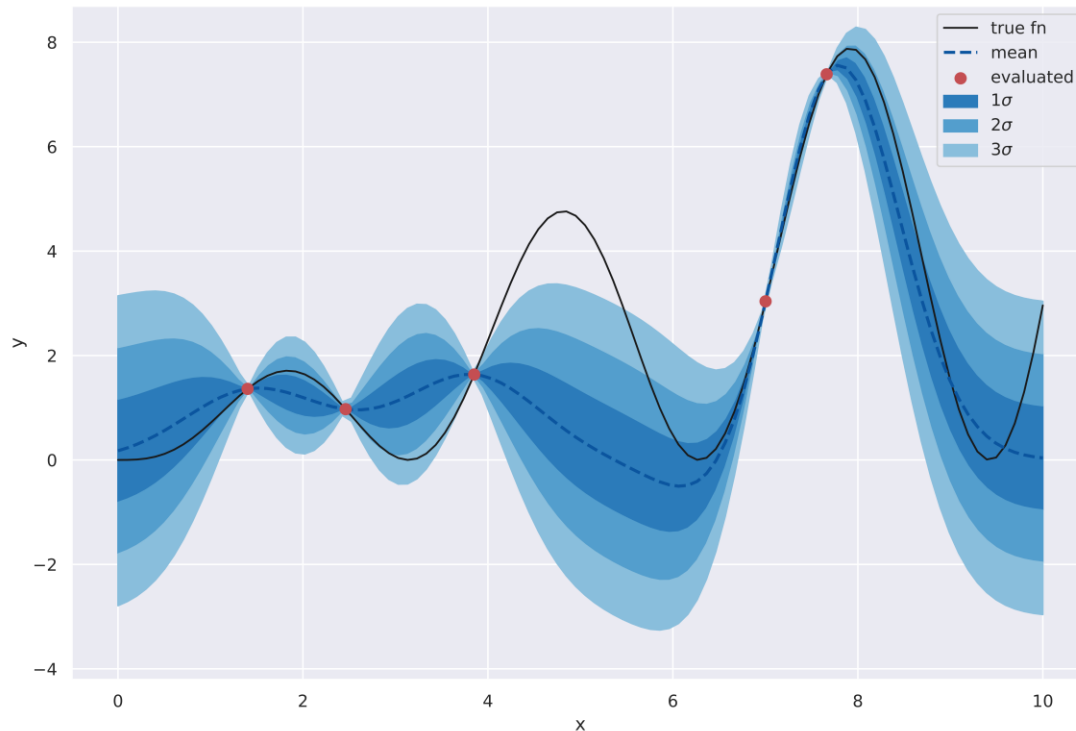
BEAM EMITTANCE



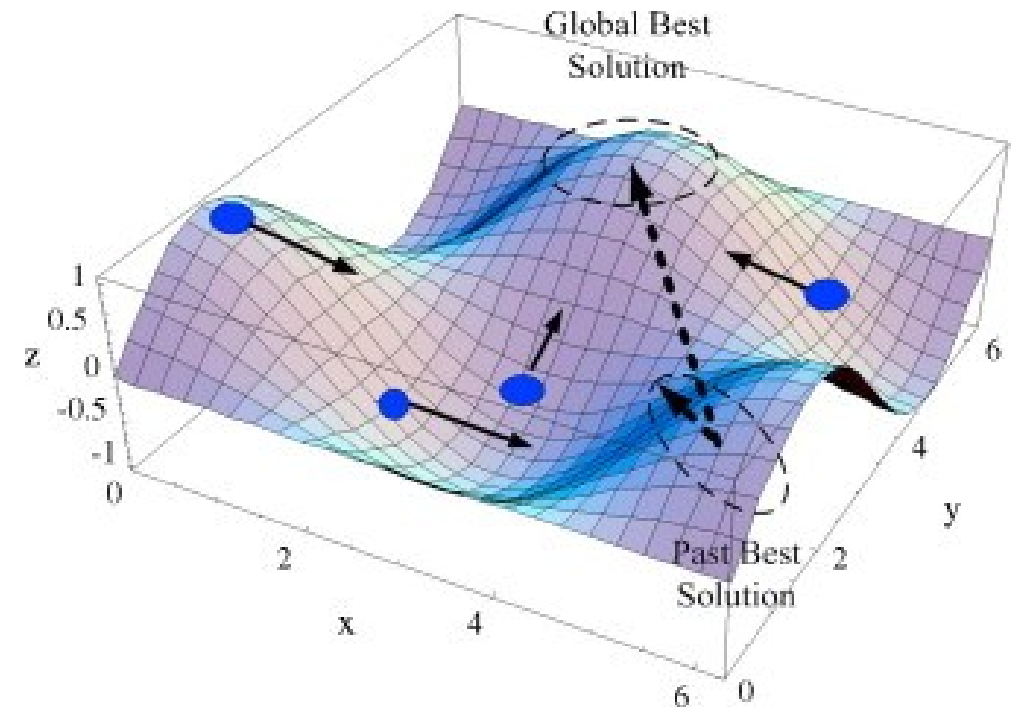


BAYESIAN OPT & PSO

BAYESIAN OPT



PSO



- Are good technique but they has «no memory», so they need to recalculate each time the next action.
- Once RL agent is trained, it can infer the action immediately.



RL PROBLEMS

Reinforcement Learning can be used to tune modules in order to optimize beam's emittance and beam transmission.

MAIN PROBLEM: Reinforcement Learning algorithms require millions of steps

- Real Accelerator is not always available and slow (5s / step)
- The existing simulators are too complex and slow (30s / step)

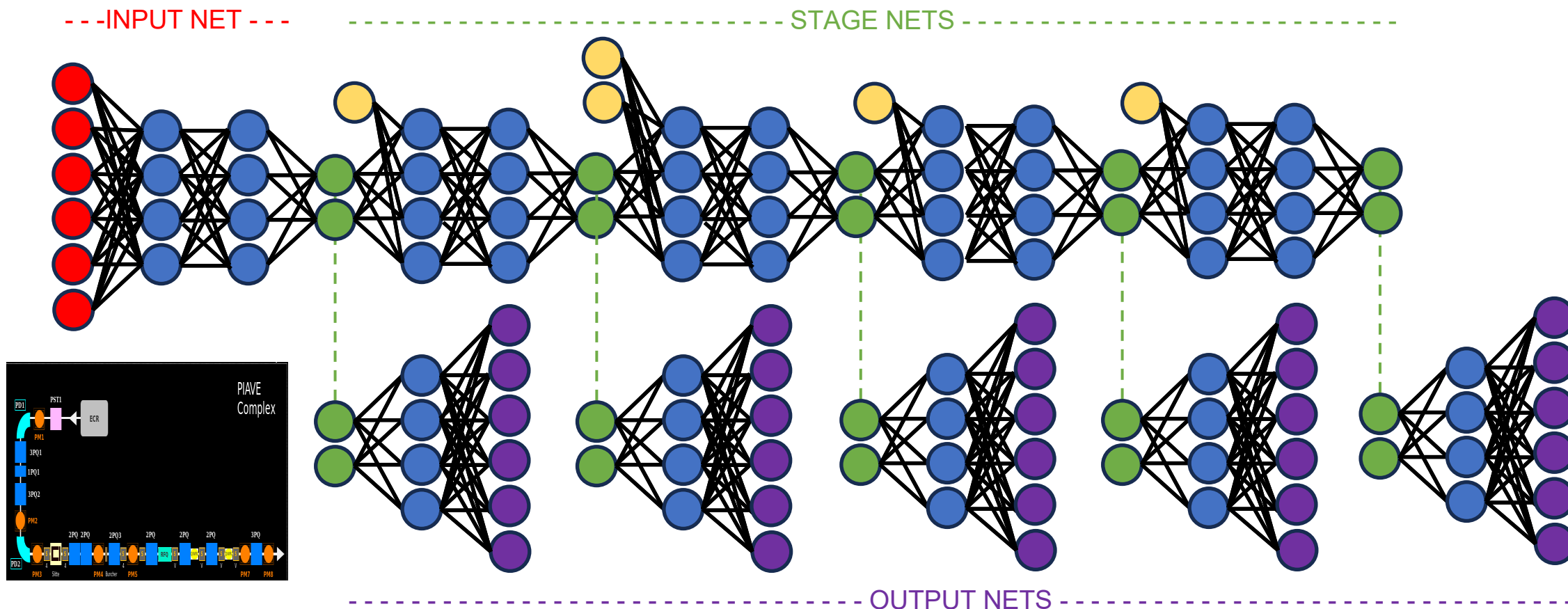
SOLUTION: CREATE A CUSTOM SIMULATOR that forecast emittance based on the module configuration



NN SIMULATOR

Train a neural (modular) network simulator

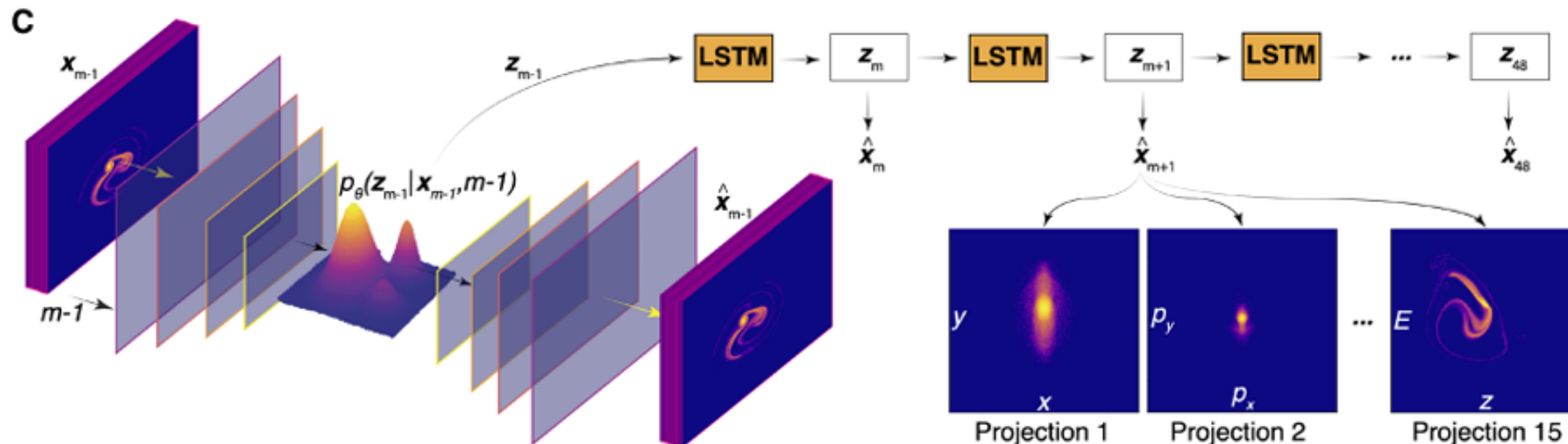
Advantages: virtual diagnostics, gradient propagation, fast inference (good for RL)



ADDITIONAL TOPIC

Conditional Variational AutoEncoder (CVAE)

Learn a low-dimensional latent space distribution in order to forecast emittance images.



$$ELBO(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z} | \mathbf{x}, m)} [\log p_{\theta}(\mathbf{x} | \mathbf{z})] - D_{KL}(q_{\phi}(\mathbf{z} | \mathbf{x}, m) || p_{\theta}(\mathbf{z} | m)).$$



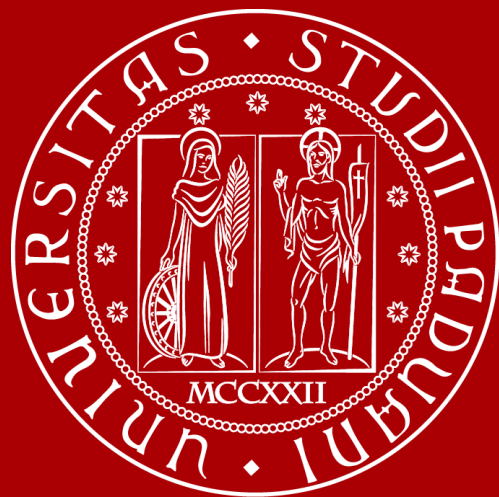
PROPOSAL

The project aims to reduce the computational complexity and improve the execution time of a physical accelerator simulator (**TraceWin**) through the development of a neural network–based surrogate model.

In the first phase, a neural simulator will be designed and trained to approximate the behavior of the original physics-based simulator. The model will integrate both data generated by **TraceWin** and experimental data collected from the real accelerator. The objective is to obtain a differentiable model that enables faster inference while maintaining sufficient accuracy in reproducing beam dynamics.

Subsequently, the neural simulator will be employed as an environment for training a **Reinforcement Learning** (RL) agent. The goal is to optimize the operational parameters of the accelerator and maximize beam quality according to predefined physical metrics exploring several algorithms.

Finally, the performance of the RL agent will be systematically compared with that of optimization algorithms currently in use, including **Bayesian Optimization** (BO), **Particle Swarm Optimization** (PSO), and gradient descent applied directly to the differentiable neural simulator. The comparison will be conducted in terms of achieved beam quality, computational efficiency, robustness, and generalization capability.



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THANKS FOR YOUR ATTENTION